⚛️ EduTrack — Frontend Development Document

Academic Management Platform

VTU Internship 2026 — Team 15

1. React Application Overview

EduTrack's frontend is built using **React 19**, a JavaScript library for building user interfaces. The application is a Single Page Application (SPA) that dynamically renders content based on the logged-in user's role without full page reloads.

The app was bootstrapped using **Create React App** which provides a pre-configured build setup with Webpack, Babel, and ESLint out of the box.

Key characteristics:

- Single Page Application (SPA)
- Hash-based routing using `window.history.pushState`
- Role-based rendering — different portals for Admin, Teacher, and Student
- Dark/Light theme support using React Context
- Fully responsive layout

2. Folder Structure

```
EduTrack-Frontend/
├── public/
│   ├── index.html
│   ├── favicon.png
│   └── edutrack-logo.png
├── src/
│   ├── assets/              # CSS files and static images
│   │   ├── App.css
│   │   ├── LandingPage.css
│   │   ├── index.css
│   │   └── edutrack-logo.png
│   ├── components/          # Reusable UI components
│   │   └── ThemeToggle.jsx
│   ├── context/             # React Context providers
│   │   └── ThemeContext.jsx
```

```
│   ├── pages/                # Full page components
│   │   ├── Dashboard.jsx     # Main dashboard (all portals)
│   │   ├── Login.jsx         # Login page
│   │   ├── Signup.jsx        # Student registration
│   │   ├── ForgotPassword.jsx
│   │   └── LandingPage.jsx
│   ├── App.js                # Root component and routing
│   ├── theme.js              # Font and theme constants
│   └── index.js              # Entry point
├── tests/
│   └── e2e/                  # Playwright E2E tests
├── .env                      # Environment variables
├── .env.example              # Template for env variables
├── package.json
└── playwright.config.js
```

3. Components and Hooks

3.1 Pages (Major Components)

COMPONENT	DESCRIPTION
`LandingPage.jsx`	Home page with features overview and navigation
`Login.jsx`	Role-based login with Student, Teacher, Admin selector
`Signup.jsx`	Student self-registration form
`ForgotPassword.jsx`	3-step OTP-based password reset
`Dashboard.jsx`	Main application — contains all portals

3.2 Dashboard Sub-Components (inside Dashboard.jsx)

COMPONENT	ROLE	DESCRIPTION
`AdminDashboard`	Admin	Overview stats
`AdminStudents`	Admin	Student CRUD management
`AdminTeachers`	Admin	Teacher CRUD management
`AdminCourses`	Admin	Course management
`StudentDashboard`	Student	Student overview

COMPONENT	ROLE	DESCRIPTION
`StudentCourses`	Student	View available courses
`StudentAttendance`	Student	View attendance per course
`StudentAssignments`	Student	View and submit assignments
`StudentMarks`	Student	View marks and CGPA
`StudentMaterials`	Student	View and download materials
`TeacherDashboard`	Teacher	Teacher overview
`TeacherCourses`	Teacher	Manage courses
`TeacherAttendance`	Teacher	Mark and view attendance
`TeacherAssignments`	Teacher	Create and manage assignments
`TeacherMarks`	Teacher	Enter and update marks
`TeacherMaterials`	Teacher	Upload course materials
`Settings`	All	Profile, password, system settings
`ChangePasswordModal`	All	Change password modal

3.3 Reusable Components

COMPONENT	DESCRIPTION
`ThemeToggle.jsx`	Dark/Light mode toggle button
`PasswordField`	Password input with show/hide toggle
`NavGlyph`	SVG icon renderer for navigation

3.4 React Hooks Used

HOOK	USAGE
`useState`	Managing local component state (forms, loading, errors)
`useEffect`	Fetching data on component mount and dependency changes
`useContext`	Accessing theme (dark/light) from ThemeContext

4. Role-Based Access Logic

EduTrack implements role-based access on the frontend by rendering completely different portals based on the logged-in user's role.

4.1 How it works

When a user logs in, the backend returns a `role` field (`admin`, `teacher`, or `student`). This role is stored in the `App.js` state and passed down to the `Dashboard` component.

```
// App.js
const [role, setRole] = useState("student");
const [user, setUser] = useState(null);

const handleLogin = (r, data) => {
  setRole(r);       // "admin", "teacher", or "student"
  setUser(data);    // user profile data
  setPage("dashboard");
};
```

4.2 Portal Rendering

```
// Dashboard.jsx
const isStudent = role === "student";
const isAdmin   = role === "admin";
const isTeacher = role === "teacher";

// Navigation items change per role
const navItems = isStudent ? STUDENT_NAV : isAdmin ? ADMIN_NAV :
TEACHER_NAV;

// Pages rendered per role
case "Settings":
  return <Settings role={role} user={user} />;
```

4.3 Navigation per Role

ROLE	NAVIGATION ITEMS
Admin	Dashboard, Students, Teachers, Courses, Settings
Teacher	Dashboard, Manage Courses, Mark Attendance, Assignments, Enter Marks, Course Materials, Settings

ROLE	NAVIGATION ITEMS
Student	Dashboard, My Courses, Attendance, Assignments, My Marks, Materials, Settings

4.4 Maintenance Mode

When maintenance mode is enabled by admin, students and teachers see an error on login:

```
"Platform is under maintenance. Please try again later."
```

Admin login is not affected by maintenance mode.

5. Modules Installed

MODULE	VERSION	PURPOSE
`react`	19.2.4	Core React library
`react-dom`	19.2.4	DOM rendering
`react-scripts`	5.0.1	Build toolchain (CRA)
`@testing-library/react`	16.3.2	Component testing utilities
`@testing-library/jest-dom`	6.9.1	Jest DOM matchers
`@testing-library/user-event`	13.5.0	User interaction simulation
`@playwright/test`	Latest	End-to-end browser testing
`web-vitals`	2.1.4	Performance metrics
`@fortawesome/fontawesome-free`	7.2.0	Icon library

*Note: EduTrack uses the native **Fetch API** for HTTP requests instead of Axios, and **window.history** for routing instead of react-router-dom, keeping the dependency footprint minimal.*

6. APIs Used

6.1 Internal REST API (EduTrack Backend)

All data operations use the EduTrack Spring Boot backend REST API.

Base URL (Local)	`http://localhost:8080`
Base URL (Production)	`https://edutrack-backend-o1ym.onrender.com`

The API URL is configured via environment variable:

```
REACT_APP_API_URL=http://localhost:8080
```

Used in frontend as:

```
const API = process.env.REACT_APP_API_URL || "http://localhost:8080";
fetch(`${API}/api/auth/login/admin`, { ... });
```

6.2 Gmail SMTP (via Backend)

OTP emails for password reset are sent through the backend using Gmail SMTP. The frontend triggers this by calling:

```
POST /api/otp/forgot-password
```

The backend handles the actual email sending — the frontend only makes the API call.

6.3 No Third-Party APIs

EduTrack does not use any third-party APIs such as:

- No payment gateway
- No map API
- No chatbot/AI API
- No social login (Google/Facebook)

All functionality is self-contained within the EduTrack backend.

7. REST API Integration

The frontend communicates with the backend exclusively through **REST APIs** using the native **Fetch API**.

7.1 Standard API Call Pattern

```
const res = await fetch(`${API}/api/auth/login/admin`, {
  method: "POST",
```

```
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify({ email, password })
});
const data = await res.json();
if (data.success) {
  // handle success
} else {
  setError(data.message);
}
```

7.2 File Upload Pattern (multipart/form-data)

```
const formData = new FormData();
formData.append("file", file);
formData.append("courseId", courseId);

const res = await fetch(`${API}/api/materials/upload`, {
  method: "POST",
  body: formData   // No Content-Type header — browser sets it
automatically
});
```

7.3 Error Handling

```
try {
  const res = await fetch(...);
  if (!res.ok) {
    setError(`Server error (${res.status})`);
    return;
  }
  const data = await res.json();
  if (!data.success) setError(data.message);
} catch {
  setError("Cannot connect to server. Make sure the backend is
running.");
}
```

7.4 API Modules Covered

MODULE	ENDPOINTS USED
Authentication	Login, Register, Change Password
OTP	Send OTP, Verify OTP, Reset Password

MODULE	ENDPOINTS USED
Students	CRUD, Status update, Bulk import
Teachers	CRUD, Status update, Bulk import
Courses	CRUD
Attendance	Mark, View, Bulk import
Marks	Enter, View, Bulk import
Assignments	CRUD
Submissions	Upload, View, Download
Course Materials	Upload, Link, Download, Delete
System Settings	Get, Update

8. Theme System

EduTrack supports **Dark and Light mode** using React Context.

```
// ThemeContext.jsx
const ThemeContext = createContext();

export function ThemeProvider({ children }) {
  const [isDark, setIsDark] = useState(true);
  const theme = isDark ? darkTheme : lightTheme;
  return (
    <ThemeContext.Provider value={{ theme, isDark, toggleTheme }}>
      {children}
    </ThemeContext.Provider>
  );
}
```

All components access the theme via:

```
const { theme: C } = useTheme();
// Use C.text, C.surface, C.teal, etc.
```

9. Environment Variables

VARIABLE	DESCRIPTION
`REACT_APP_API_URL`	Backend API base URL
`DISABLE_ESLINT_PLUGIN`	Disables ESLint during production build